Week 8: Gaussian Process Regression

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1 Brownian Motion and Integrated Brownian Motion

We studied the following model in the lectures. We have a response variable y and a single covariate x. In our example, y denotes weekly earnings and x denotes years of experience. The covariate x takes the values $0, 1, \ldots, m$ for some fixed integer m.

Our data is $(x_i, y_i), i = 1, \ldots, n$. We consider the model:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 \,\mathrm{ReLU}(x_i - 1) + \cdots + \beta_m \,\mathrm{ReLU}(x_i - (m-1)) + \epsilon_i \tag{1}$$

where $\mathrm{ReLU}(u) := \max(u, 0)$, and $\epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2)$.

We discussed Bayesian inference in this model using the prior:

$$\beta_0 \sim N(0, C), \beta_1 \sim N(0, C), \beta_2, \ldots, \beta_m \overset{\text{i.i.d}}{\sim} N(0, \tau^2) \tag{2}$$

We remarked that the resulting prior on the regression function:

$$f(x) = \beta_0 + \beta_1 x + \beta_2 \,\mathrm{ReLU}(x-1) + \cdots + \beta_m \,\mathrm{ReLU}(x-(m-1))$$

with prior (2) is very similar to:

$$f(x) = \beta_0 + \beta_1 x + \tau I(x) \tag{3}$$

where $I(x)$ is integrated Brownian motion (and $\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C)$). We shall now see why the Integrated Brownian Motion is appearing here.

1.1 Defining Brownian Motion

Before defining Brownian motion, let us recall the definition of Brownian motion. Brownian motion on $[0, M]$ (for some fixed $M > 0$) is a stochastic process $\{B(t), 0 \le t \le M\}$ characterized by the following two properties:

1. Every realization is a continuous function on $[0, M]$, and $B(0) = 0$.
2. For every fixed $t_1, \ldots, t_k \in [0, M]$, the random vector $(B(t_1), \ldots, B(t_k))$ has a multivariate normal distribution with mean zero and covariance matrix

$$\mathrm{Cov}(B(t_i), B(t_j)) = \min(t_i, t_j).$$

This definition suggests the following method for simulating $B(t)$. Construct the dense grid

$$0 = t_0 < t_1 < \cdots < t_N = M$$

on $[0, M]$ with $t_i = iM/N$. One can then simulate $(B(t_1), \ldots, B(t_N))$ from the multivariate normal distribution with mean zero and covariance matrix $\min(t_i, t_j)$.

However, this approach is computationally inefficient for large N. Sampling directly from an $N \times N$ covariance matrix typically requires a matrix factorization (such as a Cholesky decomposition), which has computational cost on the order of $O(N^3)$.

For simulations in particular, the following alternative definition is more appealing:

1. Every realization is a continuous function on $[0, M]$ with $B(0) = 0$.
2. The process has independent increments which means that $B(t_1), B(t_2) - B(t_1), \ldots, B(t_k) - B(t_{k-1})$ are independent whenever $0 \le t_1 < \cdots < t_k \le M$.
3. $B(t) - B(s) \sim N(0, t - s)$ whenever $0 \le s \le t$.

With this definition, we can simulate $B(t_1), \ldots, B(t_N)$ for $t_i = iM/N$ in the following way:

- First generate $\alpha_1, \ldots, \alpha_N \overset{\text{i.i.d}}{\sim} N(0, M/N)$
- Take $B(t_i) = \alpha_1 + \cdots + \alpha_i$ for $i = 1, \ldots, N$.

Note that in this way $\alpha_i = B(t_i) - B(t_{i-1})$. The equation $B(t_i) = \alpha_1 + \cdots + \alpha_i$ can also be written as

$$B(t) = \sum_{i=1}^{N} \alpha_i I\{t_i \le t\} \quad \text{for } t \in \{t_1, \ldots, t_N\}.$$

Because of the denseness of the grid (when N is large), we are claiming that

$$B(t) \approx \sum_{i=1}^{N} \alpha_i I\{t_i \leq t\} \quad \text{for all } t \in [0, M]. \tag{4}$$

This means that Brownian motion can be well-approximated (when N is large) by a piecewise constant process which jumps at each grid point $t_i = iM/N$ by a small amount given by $N(0, M/N)$.

We are now ready to define the Integrated Brownian Motion. This is given by:

$$I(t) = \int_0^t B(s)ds. \tag{5}$$

Using the approximation (4), we can write

$$I(t) \approx \int_0^t \left(\sum_{i=1}^{N} \alpha_i I\{t_i \leq s\} \right) ds = \sum_{i=1}^{N} \alpha_i \int_0^t I\{t_i \leq s\} ds = \sum_{i=1}^{N} \alpha_i (t - t_i)_+.$$

Therefore Integrated Brownian Motion can be approximated as:

$$I(t) \approx \sum_{i=1}^{N} \alpha_i (t - t_i)_+ \tag{6}$$

when N is large, $t_i = iM/N$ and $\alpha_i \overset{\text{i.i.d}}{\sim} N(0, M/N)$.

Using (6), the (prior) model (3) can therefore be written as:

$$f(x) \approx \beta_0 + \beta_1 x + \tau \sum_{i=1}^{N} \alpha_i \left(x - \frac{iM}{N} \right)_+$$

Here N is a large number. If we do a further coarse approximation with $N = M$, we get back the model (1) if we make the identification $\beta_i = \tau \alpha_{i-1}$ for $i = 2, \ldots, N$ and $M = m$. Therefore, the model (1) can be understood as a coarse discretization of the model (3) which stipulates that f is an Integrated Brownian Motion (scaled by τ) and added to a linear function $\beta_0 + \beta_1 x$ with the uniformative $N(0, C)$ prior on the coefficients β_0, β_1.

The prior (3) is an example of a Gaussian process prior.

2 Gaussian Process Regression

Consider the usual nonparametric regression problem where the goal is to estimate an unknown function $f : \Omega \to \mathbb{R}$ from observations $(x_1, y_1), \ldots, (x_n, y_n)$ under the model:

$$y_i = f(x_i) + \epsilon_i \quad \text{where } \epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2).$$

Here Ω denotes the domain which is a subset of $\mathbb{R}^d$ for some $d \geq 1$.

We assume that $\{f(x), x \in \Omega\}$ forms a Gaussian process with mean function $\mu(x), x \in \Omega$ and covariance function or *kernel* given by $K(x, x')$ i.e.,

$$\text{Cov}(f(x), f(x')) = K(x, x') \quad \text{for all } x, x' \in \Omega.$$

The kernel is positive semi-definite i.e., for every $N \geq 1$, distinct points $u_1, \ldots, u_N \in \Omega$, the $N \times N$ matrix with $(i, j)^{th}$ entry $K(u_i, u_j)$ is positive semi-definite. Often, this matrix will be positive definite, and hence invertible.

The mean function is usually taken to be zero. Here are some standard special cases. Here are some examples of Gaussian processes and kernels:

1. **Brownian Motion**: Here $\Omega = [0, \infty)$ and $K(s,t) = \min(s,t)$.

2. **(scaled) Brownian Motion plus constant**: Here $\Omega = [0, \infty)$ and we assume that $f(t) = \beta_0 + \tau W_t$ where $W_t \sim BM$, $\beta_0 \sim N(0, C)$ and $\tau > 0$ (and independence between β_0 and $\{W_t\}$). C will be taken to be large. Now

$$K(s,t) = C + \tau^2 \min(s,t).$$

3. **Integrated Brownian Motion**: $\Omega = [0, \infty)$ and

$$f(t) = \int_0^t W_s ds \quad \text{where } W_s \sim BM.$$

The kernel is given by

$$\begin{aligned} K(s,t) &= \text{Cov}\left(\int_0^s W_u du, \int_0^t W_v dv\right) \\ &= \int_0^s \int_0^t \text{Cov}(W_u, W_v) dv du = \int_0^s \int_0^t \min(u,v) dv du \end{aligned}$$

To simplify further, assume that $s \leq t$ so that

$$\begin{aligned} K(s,t) &= \int_0^s \left(\int_0^u v dv + \int_u^t u dv\right) du \\ &= \int_0^s \left(\frac{u^2}{2} + u(t-u)\right) du = \int_0^s \left(ut - \frac{u^2}{2}\right) du = t\frac{s^2}{2} - \frac{s^3}{6}. \end{aligned}$$

For general s and t, we have

$$K(s,t) = \frac{1}{2}\max(s,t)\left(\min(s,t)\right)^2 - \frac{1}{6}\left(\min(s,t)\right)^3.$$

4. **(scaled) Integrated Brownian Motion Plus a Linear Term**: In the IBM model, we have $f(0) = 0$ and $f'(0) = 0$. This might be an unrealistic assumption to make when f is completely unknown. In this case, a better model might be

$$f(t) = \beta_0 + \beta_1 t + \tau \int_0^t W_s ds$$

where $\beta_0, \beta_1, \{W_s\}$ are independent with W_s being Brownian motion and $\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C)$. The kernel now becomes

$$\begin{aligned} K(s,t) &= \text{Cov}\left(\beta_0 + \beta_1 s + \tau \int_0^s W_u du, \beta_0 + \beta_1 t + \tau \int_0^t W_v dv\right) \\ &= C(1 + st) + \frac{\tau^2}{2}\max(s,t)\left(\min(s,t)\right)^2 - \frac{\tau^2}{6}\left(\min(s,t)\right)^3. \end{aligned}$$

There are many other examples of kernels. Next we see some basic applications of Gaussian Processes.

3 Bayesian Numerical Analysis

3.1 Interpolation

Suppose we are given values of f at $0 \le x_1 < \cdots < x_n \le 1$: $f(x_1), \ldots, f(x_n)$. Let $x \in [0,1]$ be a new point (i.e., distinct from $x_1, \ldots, x_n$). What can we say about $f(x)$?

We will solve this problem assuming a Gaussian process prior for f with mean zero and covariance kernel $K(\cdot,\cdot)$. The answer then will be given by the conditional distribution of $f(x)$ given $f(x_1), \ldots, f(x_n)$. To compute this, we note that:

$$(f(x_1), \ldots, f(x_n), f(x))^T \sim N\left(0, \begin{pmatrix} (K(x_i,x_j))_{n\times n} & (K(x_i,x))_{n\times 1} \\ (K(x,x_i))_{1\times n} & K(x,x) \end{pmatrix}\right).$$

Using the notation $K = (K(x_i,x_j))_{n\times n}$ and $\mathbf{k} = (K(x_i,x))_{n\times 1}$, we can write the conditional distribution of $f(x)$ given $f(x_1), \ldots, f(x_n)$ as

$$f(x) \mid f(x_1), \ldots, f(x_n) \sim N\left(\mathbf{k}^T K^{-1} (f(x_1), \ldots, f(x_n))^T, K(x,x) - \mathbf{k}^T K^{-1} \mathbf{k}\right).$$

Thus the posterior mean (or mode) estimate of $f(x)$ is given by

$$\widehat{f(x)} = \mathbf{k}^T K^{-1} (f(x_1), \ldots, f(x_n))^T. \tag{7}$$

For different kernels K, the above expression behaves differently. The simplest example is that of Brownian Motion.

Example 3.1 (Brownian Motion)**.** Suppose $f(x) = \beta_0 + \tau B(x)$ where $B(x)$ is Brownian Motion, $\beta_0 \sim N(0, C)$ with $C \to \infty$ and τ is a fixed positive constant. Then the kernel is:

$$K(x_1, x_2) = C + \tau^2 \min(x_1, x_2). \tag{8}$$

In this case, (7) can be explicitly evaluated (in the limit $C \to \infty$) as

$$\widehat{f(x)} = \begin{cases} f(x_1), & x \le x_1, \\ \dfrac{x_{i+1} - x}{x_{i+1} - x_i} f(x_i) + \dfrac{x - x_i}{x_{i+1} - x_i} f(x_{i+1}), & x_i \le x \le x_{i+1}, \\ f(x_n), & x \ge x_n. \end{cases} \tag{9}$$

Exercise: Can you show why (7) with the kernel (8) leads to (9)?

The formula (9) for $\widehat{f(x)}$ performs linear interpolation between the observed points, and constant extrapolation outside the observed range. Also note that $\widehat{f(x)}$ in (9) does not depend on τ. The hyperparameter τ only affects the posterior variance of $f(x)$, and not the posterior mean.

Next example is that of the Integrated Brownian Motion prior.

Example 3.2 (Integrated Brownian Motion)**.** Suppose

$$f(x) = \beta_0 + \beta_1 x + \tau I(x), \tag{10}$$

where $I(x)$ is integrated Brownian motion and $\beta_0, \beta_1 \overset{i.i.d.}{\sim} N(0, C)$ with $C \to \infty$. In this case, the posterior mean in (7) can be evaluated explicitly and is the natural cubic spline interpolant of the observations. We will discuss more about splines later.

3.2 Integration

Suppose we are given values of f at $0 \le x_1 < \cdots < x_n \le 1$: $f(x_1), f(x_2), \ldots, f(x_n)$. What can we say about $\int_0^1 f(x)$?

Again we place a Gaussian process prior on f. By linearity, we can write

$$\begin{aligned}
&\mathbb{E}\left[\int_0^1 f(x)dx \mid f(x_1), \ldots, f(x_n)\right] \\
&= \int_0^1 \mathbb{E}\left[f(x) \mid f(x_1), \ldots, f(x_n)\right] \\
&= \int_0^1 \mathbf{k}^T K^{-1} (f(x_1), \ldots, f(x_n))^T \\
&= \left(\int_0^1 k(x_1, x)dx, \ldots, \int_0^1 k(x_n, x)dx\right)^T K^{-1} (f(x_1), \ldots, f(x_n))^T.
\end{aligned}$$

Example 3.3 (Brownian Motion)**.** For the Brownian motion, we already have the formula (9) for $\mathbb{E}[f(x) \mid f(x_1), \ldots, f(x_n)]$. So the posterior mean for $\int_0^1 f(x)dx$ is obtained by simply integrating the right hand side of (9) from 0 to 1. This leads to:

$$\begin{aligned}
&\mathbb{E}\left(\int_0^1 f(x)dx \mid f(x_1), \ldots, f(x_n)\right) \\
&= x_1 f(x_1) + \sum_{i=1}^{n-1} (x_{i+1} - x_i) \frac{f(x_i) + f(x_{i+1})}{2} + (1 - x_n) f(x_n).
\end{aligned}$$

This the trapezoidal integration rule. Therefore, with the Brownian motion prior, the posterior mean estimate of $\int_0^1 f(x)dx$ coincides with the trapezoidal rule.

Example 3.4 (Integrated Brownian Motion)**.** With the prior (10), we get a version of the Simpson integration rule.

For more on the relation between classical integration (quadrature) rules and Gaussian processes, see the book Hennig, Osborne, and Kersting (2022) or the paper Diaconis (1988).

In the next section, we shall see how to use Gaussian processes for the regression problem.

3.3 Natural Cubic Spline Interpolation

Suppose we are given observations

$$(x_1, y_1), \ldots, (x_n, y_n), \qquad x_1 < \cdots < x_n,$$

and we want to construct a function f that passes through all these points.

A natural question is:

What is the smoothest curve that passes through all the observed points?

Let us first think about what we mean by a smooth curve.

Linear interpolation. The simplest approach is linear interpolation: join each pair of successive observations by a straight line.

The resulting function is continuous, but its derivative generally jumps at the observed points. In other words, the interpolating curve has sharp corners. We would like to construct an interpolating curve that is smoother.

Cubic spline interpolation. One natural idea is to fit a cubic polynomial between each pair of successive observations.

A *cubic spline* is a function s satisfying the following properties:

1. On each interval $[x_i, x_{i+1}]$, the function s is a polynomial of degree at most three.
2. It interpolates the observations:
$$s(x_i) = y_i, \qquad i = 1, \ldots, n.$$
3. The functions s, s', and s'' are continuous throughout $[x_1, x_n]$.

The third condition ensures that the individual cubic pieces join smoothly, without sudden changes in either slope or curvature.

However, these conditions do not yet determine a unique interpolating curve. We need two additional boundary conditions.

Natural cubic splines. A *natural cubic spline* is a cubic spline satisfying the additional boundary conditions

$$\boxed{s''(x_1) = s''(x_n) = 0.}$$

Thus, the curvature vanishes at the two endpoints. Outside the observed range $[x_1, x_n]$, the natural cubic spline is extended linearly.

These conditions determine a unique interpolating function, called the *natural cubic spline interpolant*.

A variational characterization. Perhaps the most appealing characterization of the natural cubic spline interpolant is the following.

Among all sufficiently smooth functions that interpolate the observations, the natural cubic spline minimizes the total squared curvature:

$$\boxed{\begin{array}{ll} \underset{g}{\text{minimize}} & \displaystyle\int_{x_1}^{x_n} \big(g''(x)\big)^2\, dx \\ \text{subject to} & g(x_i) = y_i, \qquad i = 1, \ldots, n. \end{array}}$$

Here the minimization is over functions whose second derivatives are square-integrable.

This gives a particularly intuitive interpretation:

- The interpolation constraints force the curve to pass through all the observed points.
- The integral penalizes curvature. A function with a rapidly changing slope has a large value of this integral.
- The natural cubic spline is the interpolating curve with the smallest possible total squared curvature.

A straight line has zero curvature. However, unless all the observations lie on a straight line, no straight line can interpolate them. The natural cubic spline finds the interpolating curve that bends as little as possible, in the sense of minimizing the integral above.

Remarkable fact. In the optimization problem above, we never required the solution to be a piecewise cubic polynomial. We only asked for the smoothest interpolating function. The fact that the solution is a natural cubic spline, and hence piecewise cubic, emerges automatically from the optimization problem. Also it is remarkable that the Integrated Brownian Motion prior gives rise to this object as the posterior mean. Can you argue why?

4 Nonparametric Regression with Gaussian Processes

We will see how to perform regression using Gaussian processes. The key difference between **regression** and **interpolation** is that, in regression, the values $f(x_1), \ldots, f(x_n)$ are not observed exactly; instead, they are observed with noise.

More precisely, we are given observations $y_1, \ldots, y_n$ modeled as

$$y_i = f(x_i) + \epsilon_i, \qquad \text{where } \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

The parameter σ controls the level of noise. As in the interpolation setting, our goal is to estimate $f(x)$ at a test point x. This test point may be different from the observed inputs $x_1, \ldots, x_n$, or it may coincide with one of them.

Importantly, because the observations are noisy, it is meaningful to estimate $f(x_i)$ even at observed points. In fact, by combining information from all observations, we can often obtain a better estimate of $f(x_i)$ than the raw observation y_i. Another key difference from interpolation is that the input points $x_1, \ldots, x_n$ need not necessarily be distinct. Since each observation contains noise, having repeated measurements at the same input can help improve the overall estimate.

The solution to the regression problem is very similar to that of the interpolation problem with only one difference. The goal is to estimate $f(x)$ at the test point x given the data $(x_1, y_1), \ldots, (x_n, y_n)$. We will use the conditional distribution of $f(x)$ given $y_1, \ldots, y_n$ (we will assume that $x_1, \ldots, x_n, x$ are deterministic). To calculate this conditional distribution, first note that the marginal distribution of $(y_1, \ldots, y_n, f(x))$ is given by

$$(y_1, \ldots, y_n, f(x)) \sim N\left(0, \begin{pmatrix} (K(x_i, x_j))_{n\times n} + \sigma^2 I_n & (K(x_i, x))_{n\times 1} \\ (K(x, x_i))_{1\times n} & K(x, x) \end{pmatrix}\right).$$

Using the notation $K = (K(x_i, x_j))_{n\times n}$ and $\mathbf{k} = (K(x_i, x))_{n\times 1}$, we can write the conditional distribution of $f(x)$ given $y_1, \ldots, y_n$ as

$$f(x) \mid \text{data} \sim N\left(\mathbf{k}^T \left(K + \sigma^2 I_n\right)^{-1} y, K(x, x) - \mathbf{k}^T \left(K + \sigma^2 I_n\right)^{-1} \mathbf{k}\right).$$

Thus the posterior mean (or mode) estimate of $f(x)$ is given by

$$\widehat{f(x)} = \mathbf{k}^T \left(K + \sigma^2 I_n\right)^{-1} y, \tag{11}$$

where y is the $n \times 1$ vector with entries $y_1, \ldots, y_n$.

So the only difference between (11) and (7) is the presence of the additional $\sigma^2 I_n$ term in case of regression.

Often, we would need to estimate σ from the observed data (and also additional hyperparameters present in the kernel K). For these, the marginal likelihood of $y_1, \ldots, y_n$ is important. This is simply the multivariate normal distribution with mean vector 0 and covariance $K+\sigma^2 I_n$. We now illustrate these calculations using the integrated Brownian motion prior.

5 Implementing Regression with the IBM Prior

We now explain how to implement Gaussian process regression with an integrated Brownian motion (IBM) prior and a linear drift. There are two questions to answer. First, if the hyperparameters are known, what is our estimate of the regression function? Second, how do we learn the hyperparameters from the data? The first question is answered by the general Gaussian process posterior

formula. For the second, we integrate out the regression function and use the resulting likelihood to carry out hierarchical Bayesian inference.

We organize the calculation into steps, emphasizing what each step accomplishes. The matrix algebra gives explicit formulas, but the main idea is simple: average out the unknown quantities in an order that leaves only a one-dimensional numerical calculation.

The model and notation

Suppose that

$$y_i = f(x_i) + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0, \sigma^2), \qquad 0 \le x_1 < \cdots < x_n \le M.$$

We use the prior

$$\boxed{f(x) = \beta_0 + \beta_1 x + \sigma\gamma I(x),} \qquad \beta_0, \beta_1 \overset{\text{i.i.d.}}{\sim} \mathrm{N}(0, C),$$

where $I(x)$ is integrated Brownian motion and all the components are independent. The constant C will tend to infinity, making the prior on the intercept and slope diffuse.

Here σ is the observation-noise standard deviation, while γ controls the size of the nonlinear component relative to the noise. A small γ favors a curve close to a straight line. A larger γ allows more substantial departures from a straight line. We work with this parametrization throughout as the posterior calculations are more convenient.

Recall the IBM covariance kernel

$$k_I(s,t) = \frac{1}{2}\min(s,t)^2 \max(s,t) - \frac{1}{6}\min(s,t)^3.$$

Let $y = (y_1, \ldots, y_n)^T$ and define

$$X = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad K_I = (k_I(x_i, x_j))_{i,j=1}^n, \qquad A_\gamma = I_n + \gamma^2 K_I.$$

For a prediction input x, write

$$h_x = (1, x)^T, \qquad k_x = (k_I(x_1, x), \ldots, k_I(x_n, x))^T.$$

The matrix A_γ describes the covariance of the nonlinear component plus observation noise, after factoring out σ^2.

5.1 Step 1: Find the posterior mean for fixed hyperparameters

What this step does. We first treat γ and σ as known and ask what curve the model fits. We can apply the general GP regression formula directly.

For finite C, the prior covariance matrix of the function values at the observed inputs is

$$K_C = CXX^T + \sigma^2\gamma^2 K_I.$$

The covariance vector between these function values and $f(x)$ is

$$k_C(x) = CXh_x + \sigma^2\gamma^2 k_x.$$

Therefore the general GP posterior mean is

$$\mathbb{E}[f(x) \mid y, \gamma, \sigma] = k_C(x)^T (K_C + \sigma^2 I_n)^{-1} y. \tag{12}$$

This formula already gives an implementation for finite C. However, rather than choosing an arbitrary large value of C, we can take the diffuse-prior limit explicitly.

Define

$$D_\gamma = X^T A_\gamma^{-1} X, \qquad b_\gamma = D_\gamma^{-1} X^T A_\gamma^{-1} y.$$

Simplifying (12) with the matrix inverse identity and then taking $C \to \infty$ gives

$$\boxed{m_\gamma(x) = h_x^T b_\gamma + \gamma^2 k_x^T A_\gamma^{-1}(y - X b_\gamma).} \tag{13}$$

The formula can be understood without following the matrix simplification line by line. The first term is a fitted straight line: b_γ is the generalized least-squares estimate of its intercept and slope. The second term uses the IBM covariance to estimate the nonlinear structure remaining in the residuals $y - X b_\gamma$.

A useful consequence is that *the conditional posterior mean depends on γ, but not on σ*. Thus, once we know γ, we know the fitted mean curve. The noise scale still matters for posterior uncertainty.

For example, as $\gamma \to 0$, $A_\gamma \to I_n$, the nonlinear correction vanishes, and the posterior mean approaches the ordinary least-squares line. Increasing γ permits a more flexible fit. This identifies γ as the smoothing hyperparameter that we need to learn.

5.2 Step 2: Obtain a likelihood involving only the hyperparameters

What this step does. To learn γ and σ, we need to measure how well each pair explains the observed data. We do this by averaging over the regression function under its prior, rather than selecting one particular function.

For finite C, the function values and observation errors are independent Gaussian vectors, so their sum has distribution

$$y \mid \gamma, \sigma \sim \mathrm{N}(0, CXX^T + \sigma^2 A_\gamma).$$

Hence the marginal likelihood is

$$p_C(y \mid \gamma, \sigma) \propto |CXX^T + \sigma^2 A_\gamma|^{-1/2} \exp\left\{ -\frac{1}{2} y^T (CXX^T + \sigma^2 A_\gamma)^{-1} y \right\}.$$

We again take $C \to \infty$. In this limit, the drift prior becomes flat, so the resulting integrated likelihood is defined up to a common multiplicative constant. More precisely, we remove a factor depending on C but not on γ or σ before taking the limit.

Let

$$R_\gamma = (y - X b_\gamma)^T A_\gamma^{-1} (y - X b_\gamma). \tag{14}$$

This is a weighted residual sum of squares after fitting the linear drift. The limiting integrated likelihood is

$$\boxed{L(\gamma, \sigma; y) \propto \sigma^{-(n-2)} |A_\gamma|^{-1/2} |D_\gamma|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2} \right).} \tag{15}$$

An equivalent way to obtain this expression is to start with $y \mid \beta, \gamma, \sigma \sim \mathrm{N}(X\beta, \sigma^2 A_\gamma)$ and integrate the two drift coefficients against a flat prior. Both routes give the same expression.

There are three features to notice. The exponential term measures fit through R_γ. The determinant factors arise because we have integrated over the unknown regression quantities; they are part of the likelihood and must be retained. Finally, the power of σ is $n-2$, reflecting the two fitted drift coefficients.

We have now eliminated the entire regression function from the hyperparameter calculation. Only γ and σ remain unknown in (15).

5.3 Step 3: Learn the noise variance conditional on the smoothing parameter

What this step does. We now place priors on the hyperparameters. A convenient scale prior makes the conditional posterior of the noise variance available in closed form, so no numerical grid is needed for it.

Take independent priors

$$p(\sigma) \propto \frac{1}{\sigma}, \qquad \sigma > 0, \qquad p(\gamma) \propto \frac{1}{\gamma}\,\mathbf{1}\{a \le \gamma \le b\}, \qquad 0 < a < b < \infty.$$

The latter is a uniform prior on $\log \gamma$ over a specified finite interval. The bounds are kept finite: extending $1/\gamma$ all the way to zero would make the posterior improper in this model. We assume $n > 2$ and that the observed responses are not exactly a linear function of the inputs, so $R_\gamma > 0$.

Multiplying the likelihood by the priors gives

$$p(\gamma, \sigma \mid y) \propto p(\gamma)\sigma^{-(n-1)}|A_\gamma|^{-1/2}|D_\gamma|^{-1/2}\exp\left(-\frac{R_\gamma}{2\sigma^2}\right).$$

For a fixed γ, everything involving only γ is a constant. Thus the dependence on σ is simply

$$p(\sigma \mid y, \gamma) \propto \sigma^{-(n-1)}\exp\left(-\frac{R_\gamma}{2\sigma^2}\right).$$

Writing this as a density for $v = \sigma^2$ gives

$$p(v \mid y, \gamma) \propto v^{-(n-2)/2-1}\exp\left(-\frac{R_\gamma}{2v}\right).$$

We recognize an inverse-gamma density:

$$\boxed{\sigma^2 \mid y, \gamma \sim \text{InverseGamma}\left(\frac{n-2}{2}, \frac{R_\gamma}{2}\right).} \tag{16}$$

Our convention is that an InverseGamma(α, β) density is proportional to $v^{-\alpha-1}e^{-\beta/v}$ for $v > 0$. Equivalently, $1/\sigma^2$ is Gamma with shape $(n-2)/2$ and rate $R_\gamma/2$.

The conceptual point is that, once γ is specified, learning the noise variance requires no optimization or numerical integration. Its posterior distribution is known explicitly, and we can compute posterior summaries or generate draws using a standard distribution. It is the variance σ^2 that has the inverse-gamma posterior.

5.4 Step 4: Integrate out the noise scale to obtain the posterior of gamma

What this step does. Rather than substituting a single estimate of the noise variance, we average over all its possible values. The inverse-gamma calculation makes this integration explicit and leaves a posterior involving only γ.

The integral over σ is

$$\int_0^\infty \sigma^{-(n-1)} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right) d\sigma \propto R_\gamma^{-(n-2)/2},$$

where the omitted factor depends only on n. It follows that

$$\boxed{p(\gamma \mid y) \propto p(\gamma)|A_\gamma|^{-1/2}|D_\gamma|^{-1/2}R_\gamma^{-(n-2)/2}.} \tag{17}$$

For the log-uniform prior, $p(\gamma)$ contributes the factor $1/\gamma$ on $[a, b]$.

This posterior need not belong to a named family, but it is only one-dimensional. We can evaluate it on a sufficiently fine grid and normalize the resulting weights. Thus the only numerical integration in the hyperparameter analysis concerns one scalar parameter.

5.5 Step 5: Combine the pieces to estimate the curve and its uncertainty

What this step does. We now combine the conditional GP posterior with the posterior of the hyperparameters. This propagates uncertainty about smoothness and noise into our inference about the regression function.

For a posterior mean curve, the calculation is particularly simple. Since $m_\gamma(x)$ does not depend on σ,

$$\boxed{\mathbb{E}[f(x) \mid y] = \int m_\gamma(x)p(\gamma \mid y)\, d\gamma.}$$

On the grid, this is just the weighted average of the curves $m_\gamma(x)$. Thus full Bayesian estimation averages over different amounts of smoothing instead of requiring a single choice.

To generate posterior curves and credible intervals, proceed in the following order:

1. Draw γ using the posterior grid weights from Step 4.
2. Draw σ^2 from the inverse-gamma distribution in Step 3.
3. Given these hyperparameters, draw the function values at the desired inputs from their joint Gaussian posterior.

Repeating these steps gives draws from the full posterior, up to the grid approximation. Pointwise quantiles give credible intervals for $f(x)$; the draws can also be used to study other features of the curve.

5.6 Automatic selection of the amount of smoothing

The parameter γ plays the role of a tuning parameter: small values favor a nearly linear fit, while larger values allow more nonlinear variation. The hierarchical Bayesian calculation learns this parameter from the data through $p(\gamma \mid y)$.

The marginal likelihood averages the likelihood over curves allowed by each value of γ. If γ is too small to explain the nonlinear pattern in the data, this average can be small. A very large γ allows a broad collection of curves, but only a small fraction may explain the observed data well. The determinant factors and residual term in (17) together reflect this balance. The posterior therefore weighs flexibility against how much prior probability is assigned to curves compatible with the data; it does not simply reward the most flexible fit.

We can summarize the posterior of γ by a point estimate and use the corresponding fitted curve. Alternatively, full Bayesian inference averages over γ, accounting for uncertainty about the appropriate smoothness. *No cross-validation or training–validation split is needed for this tuning procedure: the marginal likelihood uses the entire dataset.* The prior on γ remains part of the model, so the procedure learns smoothness within the range and prior we have specified.

5.7 Main Points for Reading

1. **What are we trying to learn?**

   There are two questions: what is the regression function, and how much smoothing should we use? We first calculate the posterior of the function for fixed hyperparameters. We then learn the hyperparameters from the data.

   Our prior is

   $$f(x) = \beta_0 + \beta_1 x + \sigma\gamma I(x).$$

   Here σ is the noise standard deviation, while γ controls the size of the nonlinear component relative to the noise. Small γ favors a nearly straight line; larger γ allows greater departures from linearity.

2. **Step 1: Apply the general GP posterior formula.**

   For fixed γ and σ, this is Gaussian process regression with a particular covariance kernel. We substitute that kernel into the general posterior mean formula.

   The constant C is the prior variance of the intercept and slope. Taking $C \to \infty$ removes their shrinkage toward zero. The limiting posterior mean is

   $$m_\gamma(x) = h_x^T b_\gamma + \gamma^2 k_x^T A_\gamma^{-1}(y - X b_\gamma),$$

   where

   $$A_\gamma = I_n + \gamma^2 K_I, \qquad b_\gamma = (X^T A_\gamma^{-1} X)^{-1} X^T A_\gamma^{-1} y.$$

   The interpretation is:

   $$\text{fitted curve} = \text{fitted linear drift} + \text{smooth correction to the residuals.}$$

   We need not follow every matrix simplification to understand what the formula does.

   Notice that this mean depends on γ, but not on σ. Thus γ determines the fitted mean curve, while σ still affects posterior uncertainty. As $\gamma \to 0$, the nonlinear correction disappears and we recover the least-squares line.

3. **Step 2: Obtain a likelihood for the hyperparameters.**

   To learn γ and σ, we integrate out the regression function, including the intercept and slope. We are averaging over possible curves rather than choosing one particular curve.

   For finite C, the observations have a multivariate Gaussian distribution:

   $$y \mid \gamma, \sigma \sim N(0, CXX^T + \sigma^2 A_\gamma).$$

   Taking the diffuse-prior limit, after removing a factor independent of the hyperparameters, gives

   $$L(\gamma, \sigma; y) \propto \sigma^{-(n-2)} |A_\gamma|^{-1/2} |D_\gamma|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right),$$

where

$$D_\gamma = X^T A_\gamma^{-1} X, \qquad R_\gamma = (y - Xb_\gamma)^T A_\gamma^{-1}(y - Xb_\gamma).$$

The quantity R_γ is a weighted residual sum of squares. The determinant factors arise from integrating over the unknown regression quantities and must be retained. The power $n-2$ reflects the two fitted drift coefficients.

We now have an expression involving only the two hyperparameters.

4. **Step 3: Recognize a closed-form posterior for the noise variance.**

   Use the prior

   $$p(\sigma) \propto \frac{1}{\sigma}, \qquad \sigma > 0.$$

   For a fixed γ, multiplying this prior by the likelihood gives

   $$p(\sigma \mid y, \gamma) \propto \sigma^{-(n-1)} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right).$$

   As a density for σ^2, this is an inverse-gamma distribution:

   $$\boxed{\sigma^2 \mid y, \gamma \sim \text{InverseGamma}\left(\frac{n-2}{2}, \frac{R_\gamma}{2}\right).}$$

   Here the inverse-gamma density with parameters (α, β) is proportional to $v^{-\alpha-1}\exp(-\beta/v)$ for $v > 0$. We assume $n > 2$ and $R_\gamma > 0$.

   The important point is computational: conditional on γ, the posterior of the noise variance is known explicitly. No numerical grid or optimization is needed for this step.

5. **Step 4: Integrate out the noise scale and learn gamma.**

   Instead of plugging in one estimate of σ, we average over its possible values. The inverse-gamma calculation allows this integration to be done explicitly:

   $$\boxed{p(\gamma \mid y) \propto p(\gamma)|A_\gamma|^{-1/2}|D_\gamma|^{-1/2}R_\gamma^{-(n-2)/2}.}$$

   For example, we can use a uniform prior on $\log\gamma$ over a specified finite interval:

   $$p(\gamma) \propto \frac{1}{\gamma}\,\mathbf{1}\{a \le \gamma \le b\}, \qquad 0 < a < b < \infty.$$

   The finite bounds are part of the prior; extending $1/\gamma$ all the way to zero would give an improper posterior here.

   The posterior of γ need not be a named distribution. Nevertheless, it is only one-dimensional, so we can evaluate it on a grid and normalize the weights. On a uniformly spaced grid in $\log\gamma$, the Jacobian cancels the $1/\gamma$ prior factor.

6. **Step 5: Combine the results to infer the curve.**

   For a posterior mean curve, average the conditional mean curves using the posterior distribution of γ:

   $$\mathbb{E}[f(x) \mid y] = \int m_\gamma(x) p(\gamma \mid y)\, d\gamma.$$

On a grid, this is simply a weighted average.

For posterior draws, proceed in the following order:

$$\gamma \mid y, \qquad \sigma^2 \mid y, \gamma, \qquad f \mid y, \gamma, \sigma.$$

First sample γ from the grid weights, then sample σ^2 from its inverse-gamma posterior, and finally sample the function values jointly from their Gaussian posterior. These draws incorporate uncertainty about both the noise level and the amount of smoothing.

7. **Why does this select the tuning parameter automatically?**

   The parameter γ determines how flexible the fitted curve can be. Its posterior tells us which levels of flexibility are supported by the data.

   The marginal likelihood averages the likelihood over curves allowed by each γ. Too little flexibility may fail to explain the observed pattern. Very large flexibility can spread prior probability over many curves that do not explain the observed data well. The posterior balances these effects.

   No cross-validation or training–validation split is needed for this tuning calculation: all observations contribute to the marginal likelihood. We can select a representative γ, or average over its posterior to account for uncertainty about the appropriate amount of smoothing.

5.8 Efficient Computation Using the Markov Structure of Integrated Brownian Motion

In our implementation of Gaussian process regression, we used the matrix

$$A_\gamma = I_n + \gamma^2 K_I,$$

where K_I is the $n \times n$ covariance matrix of integrated Brownian motion. Direct computation using this matrix costs $O(n^3)$ operations. This was one reason for restricting our numerical demonstration to a random sample of 500 observations.

However, for the integrated Brownian motion prior, there is a much more efficient approach that avoids this large matrix computation altogether.

Recall our prior

$$f(x) = \beta_0 + \beta_1 x + \sigma\gamma I(x), \qquad I(x) = \int_0^x B(t)\,dt.$$

The key observation is that although $f(x)$ itself is not a Markov process, the two-dimensional process

$$Z(x) = \begin{pmatrix} f(x) \\ f'(x) \end{pmatrix}$$

is Markov.

Indeed, for $h > 0$, the state evolves according to

$$Z(x+h) = \underbrace{\begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}}_{F_h} Z(x) + \eta_h,$$

where η_h is a mean-zero Gaussian vector with covariance

$$\boxed{Q_h = \sigma^2\gamma^2 \begin{pmatrix} h^3/3 & h^2/2 \\ h^2/2 & h \end{pmatrix}.}$$

Moreover, the innovations over disjoint intervals are independent.

The observations can be written as

$$y_i = \begin{pmatrix} 1 & 0 \end{pmatrix} Z(x_i) + \epsilon_i, \qquad \epsilon_i \sim N(0, \sigma^2).$$

Thus our Gaussian process regression model is exactly equivalent to a two-dimensional linear Gaussian state-space model. The diffuse prior on (β_0, β_1) corresponds to a diffuse initialization of the state.

We can now perform inference using two familiar algorithms:

1. **Kalman filtering:** Move forward through the observations, updating the Gaussian distribution of $Z(x_i)$ as each new observation arrives.

2. **Kalman smoothing:** Move backward through the observations to compute the posterior distributions of $Z(x_i)$ conditional on the entire dataset.

These calculations involve only matrices of size 2×2, regardless of the number of observations. Since each observation requires a constant amount of computation, the total complexity is $O(n)$.

An additional advantage is that the Kalman filter computes the marginal likelihood sequentially through the one-step predictive distributions:

$$p(y \mid \gamma, \sigma) = \prod_{i=1}^{n} p(y_i \mid y_1, \ldots, y_{i-1}, \gamma, \sigma).$$

Consequently, the hyperparameter inference discussed earlier can also be implemented without forming the $n \times n$ kernel matrix. The same diffuse-prior integrated likelihood can be obtained using a diffuse Kalman filter.

In particular, we can evaluate the posterior of γ on a grid, sample σ^2 conditionally, and use Kalman smoothing to obtain posterior means and credible intervals for the regression function.

Main takeaway. The apparent $O(n^3)$ computational cost is not intrinsic to Gaussian process regression with an integrated Brownian motion prior. By exploiting its Markov structure, exact posterior inference can be carried out in $O(n)$ time for fixed hyperparameters. This makes it possible to analyze substantially larger datasets without subsampling.

For a general discussion of this approach, see Chapter 5 of Hennig, Osborne, and Kersting, *Probabilistic Numerics*, particularly the sections on linear Gaussian state-space models, stochastic differential equations, and Kalman filtering and smoothing.

5.9 Detailed Derivations of the Posterior Formulas

This subsection supplies the calculations omitted from the main discussion. It is not needed for a first reading, but it allows the reader to verify all the posterior formulas directly. Throughout, $n > 2$, the two columns of X are linearly independent, and $R_\gamma > 0$. We use the notation

$$A = A_\gamma = I_n + \gamma^2 K_I, \quad W = A^{-1}, \quad D = X^T W X, \quad b = D^{-1} X^T W y, \quad P = W - W X D^{-1} X^T W.$$

Thus P is symmetric, $PX = 0$, and

$$R_\gamma = y^T P y = (y - Xb)^T W (y - Xb) = y^T W y - y^T W X D^{-1} X^T W y. \tag{18}$$

1. Taking the diffuse-prior limit of the posterior mean. Write $\beta = (\beta_0, \beta_1)^T$ and $f(x) = h_x^T\beta + \sigma\gamma I(x)$. After integrating out the IBM component, the conditional data model is

$$y \mid \beta, \gamma, \sigma \sim N(X\beta, \sigma^2 A), \qquad \beta \sim N(0, CI_2).$$

The usual normal–normal calculation (completing the square in β) gives

$$V_C = \left(C^{-1}I_2 + \sigma^{-2}X^TWX\right)^{-1}, \tag{19}$$

$$b_C = V_C\sigma^{-2}X^TWy = \left(\frac{\sigma^2}{C}I_2 + D\right)^{-1} X^TWy. \tag{20}$$

In particular, $b_C \to b$ and $V_C \to \sigma^2 D^{-1}$ as $C \to \infty$. Given β, the Gaussian conditioning formula for $I(x)$ and $y - X\beta$ yields

$$\mathbb{E}[\sigma\gamma I(x) \mid y, \beta, \gamma, \sigma] = \gamma^2 k_x^T W(y - X\beta).$$

The tower property now shows that, for finite C,

$$\mathbb{E}[f(x) \mid y, \gamma, \sigma] = h_x^T b_C + \gamma^2 k_x^T W(y - Xb_C).$$

Taking $C \to \infty$ proves

$$\boxed{m_\gamma(x) = h_x^T b + \gamma^2 k_x^T W(y - Xb).} \tag{21}$$

Notice that σ cancels: the mean depends on γ only. For completeness, the same limit follows directly from the Woodbury identity:

$$\begin{aligned}&(\sigma^2 A + CXX^T)^{-1}\\ &\quad = \sigma^{-2}W - \sigma^{-2}WX\left(\frac{\sigma^2}{C}I_2 + D\right)^{-1}X^TW,\end{aligned}$$

and the identity

$$CX^T(\sigma^2 A + CXX^T)^{-1} = \left(\frac{\sigma^2}{C}I_2 + D\right)^{-1}X^TW.$$

Multiplying by the finite-C cross-covariance $CXh_x + \sigma^2\gamma^2 k_x$ produces the same expression.

2. Deriving the marginal likelihood and its determinant terms. For fixed β, the Gaussian density of y is

$$\begin{aligned}p(y \mid \beta, \gamma, \sigma) = (2\pi)^{-n/2}\sigma^{-n}|A|^{-1/2}&\\ \times \exp\left\{-\frac{1}{2\sigma^2}(y - X\beta)^TW(y - X\beta)\right\}&.\end{aligned}$$

Expand its quadratic form around b:

$$\begin{aligned}(y - X\beta)^TW(y - X\beta) &= y^TWy - 2\beta^TX^TWy + \beta^TD\beta\\ &= (\beta - b)^TD(\beta - b) + R_\gamma.\end{aligned}$$

The cross term vanishes because $Db = X^T W y$. In the limit of flat drift priors, integrate over $\beta \in \mathbb{R}^2$:

$$\begin{aligned} L(\gamma, \sigma; y) &\propto \sigma^{-n}|A|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right) \int_{\mathbb{R}^2} \exp\left\{-\frac{(\beta - b)^T D(\beta - b)}{2\sigma^2}\right\} d\beta \\ &= \sigma^{-n}|A|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right) (2\pi\sigma^2)^{2/2}|D|^{-1/2} \\ &\propto \boxed{\sigma^{-(n-2)}|A|^{-1/2}|D|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right)}. \end{aligned}$$

The symbol $\propto$ omits only constants independent of (γ, σ). This flat-prior integral agrees with the $C \to \infty$ limit of the proper-prior calculation, after removing the factor depending on C alone. Indeed, the matrix determinant lemma gives

$$\begin{aligned} |\sigma^2 A + CXX^T| &= |\sigma^2 A|\,|I_2 + (C/\sigma^2)D| \\ &= \sigma^{2n}|A|\,(C/\sigma^2)^2|D + (\sigma^2/C)I_2| \\ &= C^2\sigma^{2(n-2)}|A||D + (\sigma^2/C)I_2|. \end{aligned}$$

Thus the factor C^{-1} in the square root of the Gaussian density may be discarded before passing to the limit. The Woodbury inverse above converges to $\sigma^{-2}P$, so the exponent converges to $-R_\gamma/(2\sigma^2)$ as required.

3. Posterior distribution of the variance. With $p(\sigma) \propto 1/\sigma$, conditioning on γ yields

$$p(\sigma \mid y, \gamma) \propto \sigma^{-(n-1)} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right), \qquad \sigma > 0.$$

Put $v = \sigma^2$. Because $\sigma = \sqrt{v}$ and $|d\sigma/dv| = (2\sqrt{v})^{-1}$, the density transformation is

$$\begin{aligned} p(v \mid y, \gamma) &\propto v^{-(n-1)/2} \exp\left(-\frac{R_\gamma}{2v}\right) v^{-1/2} \\ &= v^{-n/2} \exp\left(-\frac{R_\gamma}{2v}\right). \end{aligned}$$

Comparing with the inverse-gamma density $\beta^\alpha \Gamma(\alpha)^{-1} v^{-\alpha-1} e^{-\beta/v}$, we identify $\alpha = (n-2)/2$, $\beta = R_\gamma/2$, and obtain

$$\boxed{\sigma^2 \mid y, \gamma \sim \text{InverseGamma}\left(\frac{n-2}{2}, \frac{R_\gamma}{2}\right).}$$

4. Integrating out σ and obtaining $p(\gamma \mid y)$. The joint posterior under the independent prior $p(\gamma)$ is

$$p(\gamma, \sigma \mid y) \propto p(\gamma)\sigma^{-(n-1)}|A|^{-1/2}|D|^{-1/2} \exp\left(-\frac{R_\gamma}{2\sigma^2}\right).$$

Set $\alpha = (n-2)/2$ and substitute $u = R_\gamma/(2\sigma^2)$. Then $\sigma = (R_\gamma/(2u))^{1/2}$, whence

$$d\sigma = -\frac{1}{2}(R_\gamma/2)^{1/2} u^{-3/2}\, du.$$

Consequently,

$$\int_0^\infty \sigma^{-(n-1)} e^{-R_\gamma/(2\sigma^2)}\, d\sigma = \frac{1}{2}(R_\gamma/2)^{-\alpha} \int_0^\infty u^{\alpha-1} e^{-u}\, du$$
$$= \frac{1}{2}\Gamma(\alpha)(R_\gamma/2)^{-\alpha}.$$

The gamma-function factor is independent of γ. We therefore get

$$\boxed{p(\gamma \mid y) \propto p(\gamma)|A_\gamma|^{-1/2}|D_\gamma|^{-1/2}R_\gamma^{-(n-2)/2}.} \tag{22}$$

For the prior $p(\gamma) \propto \gamma^{-1}\mathbf{1}\{a \le \gamma \le b\}$, put $\eta = \log\gamma$. Since $d\gamma/d\eta = e^\eta = \gamma$,

$$p(\eta \mid y) = p(e^\eta \mid y)e^\eta \propto |A_{e^\eta}|^{-1/2}|D_{e^\eta}|^{-1/2}R_{e^\eta}^{-(n-2)/2}, \quad \log a \le \eta \le \log b.$$

Thus, on a uniform grid in η, one can normalize log weights $-\frac{1}{2}\log|A_\gamma| - \frac{1}{2}\log|D_\gamma| - \frac{n-2}{2}\log R_\gamma$ without an extra Jacobian.

5. Deriving the conditional posterior covariance. Given β, the joint Gaussian vector $(I(x), I(z), y - X\beta)$ shows that the covariance of $f(x)$ and $f(z)$, conditional on y, β, γ, σ, is

$$\sigma^2\{\gamma^2 k_I(x,z) - \gamma^4 k_x^T W k_z\}. \tag{23}$$

This covariance does not depend on β. Also, the conditional mean is

$$\mathbb{E}[f(x) \mid y, \beta, \gamma, \sigma] = h_x^T\beta + \gamma^2 k_x^T W(y - X\beta)$$
$$= \gamma^2 k_x^T W y + d_\gamma(x)^T\beta,$$

where $d_\gamma(x) = h_x - \gamma^2 X^T W k_x$. Since the flat-prior posterior is

$$\beta \mid y, \gamma, \sigma \sim N(b, \sigma^2 D^{-1}),$$

the law of total covariance adds $\sigma^2 d_\gamma(x)^T D^{-1} d_\gamma(z)$ to (23). Thus

$$\boxed{\begin{aligned}&\mathrm{Cov}(f(x), f(z) \mid y, \gamma, \sigma)\\ &\quad = \sigma^2\big[\gamma^2 k_I(x,z) - \gamma^4 k_x^T A_\gamma^{-1} k_z + d_\gamma(x)^T D_\gamma^{-1} d_\gamma(z)\big].\end{aligned}} \tag{24}$$

This derivation explains why the uncertainty in the two drift coefficients must be included when constructing credible bands. To sample a posterior curve at points $u_1, \ldots, u_q$, use the multivariate normal distribution with means $m_\gamma(u_j)$ and covariances given by (24). Drawing γ from (22) and σ^2 from the inverse-gamma posterior first produces samples from the full hierarchical posterior.

6. Checking the two-dimensional Markov representation. Because $I'(x) = B(x)$, the state is

$$Z(x) = \begin{pmatrix} f(x) \\ f'(x) \end{pmatrix} = \begin{pmatrix} \beta_0 + \beta_1 x + \sigma\gamma I(x) \\ \beta_1 + \sigma\gamma B(x) \end{pmatrix}.$$

Over an interval of length h, Brownian independent increments imply

$$f'(x+h) = f'(x) + \sigma\gamma\{B(x+h) - B(x)\}$$

and, after integrating the derivative,

$$f(x+h) = f(x) + hf'(x) + \sigma\gamma \int_0^h \{B(x+t) - B(x)\}\, dt.$$

Hence

$$Z(x+h) = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix} Z(x) + \sigma\gamma \begin{pmatrix} \int_0^h W(t)\, dt \\ W(h) \end{pmatrix}, \qquad W(t) = B(x+t) - B(x).$$

The noise vector is independent of $Z(x)$ and of innovations on disjoint intervals. Its covariance follows from $\mathrm{Cov}(W(s), W(t)) = \min(s,t)$:

$$\begin{aligned} \mathrm{Var}\left(\int_0^h W(t)\, dt \right) &= \int_0^h \int_0^h \min(s,t)\, ds\, dt = \frac{h^3}{3}, \\ \mathrm{Cov}\left(\int_0^h W(t)\, dt, W(h) \right) &= \int_0^h t\, dt = \frac{h^2}{2}, \\ \mathrm{Var}(W(h)) &= h. \end{aligned}$$

Therefore the state transition covariance is exactly

$$Q_h = \sigma^2\gamma^2 \begin{pmatrix} h^3/3 & h^2/2 \\ h^2/2 & h \end{pmatrix}.$$

With $Z(0) = (\beta_0, \beta_1)^T$, a diffuse Gaussian initialization $N(0, CI_2)$ and the observation equation $y_i = (1,0)Z(x_i) + \epsilon_i$, this state-space model is exactly the original IBM-plus-drift GP prior. Kalman filtering and smoothing use only fixed-size 2×2 matrices at each observation. For fixed (γ, σ) this gives $O(n)$ inference; the same state-space formulation, with a correctly implemented diffuse initialization, also gives the integrated likelihood for hyperparameter inference.

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